IEICE TRANS. INF. & SYST., VOL.E90–D, NO.7 JULY 2007

PAPER

Zero-Anaphora Resolution in Chinese Using Maximum Entropy

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SUMMARY In this paper, we propose a learning classifier based on maximum entropy (ME) for resolving zero-anaphora in Chinese text. Besides regular grammatical, lexical, positional and semantic features motivated by previous research on anaphora resolution, we develop two innovative Web-based features for extracting additional semantic information from the Web. The values of the two features can be obtained easily by querying the Web using some patterns. Our study shows that our machine learning approach is able to achieve an accuracy comparable to that of state-of-the-art systems. The Web as a knowledge source can be incorporated effectively into the ME learning framework and significantly improves the performance of our approach.

key words: *zero-anaphora resolution, Web-based features, maximum entropy, classifier*

1. Introduction

In many natural languages, grammatical components that can be understood contextually by a reader are frequently unexpressed for discourse fluency. This is particularly true in many Asian languages [1], such as Chinese, Japanese and Korean, where anaphoric expression is frequently eliminated. This phenomenon is called zero-anaphora (ZA) in Natural Language Processing (NLP) [2].

For example, in example (1)* below, ϕ_1 in the subject position refers to the subject "jie jie (my sister)" in the first sentence**, and it can be recovered as "ta (she)". ϕ_2 in example (2) refers to the object of the preposition phrase, "xi zi (the matting)", and it can be recovered as "ta (it)". In example (3), the grammatical role of ϕ_3 is the object. It refers to "zhe ge wen ti" (the problem) and can be recovered as "ta (it)". It is important to emphasize that in our work we only consider the inter-sentential case where the antecedent and ZA are located in different sentences, such as (1), (2) and (3). We do not take the intra-sentential case and the exophora case into account. In the intra-sentential case, the antecedent and ZA exist in the same sentence, such as example (4), in which the antecedent is the subject of the same sentence of ϕ_4. In the exophora case, the antecedent does not appear linguistically in the text because it can be understood by assumed shared knowledge between the writer and the reader, such as ϕ_5 in example (5).

(1) [姐姐]$_1$来看我, ϕ_1给了我几个苹果。
jie jie lai kan wo, (ta) gei le wo ji ge ping guo.
My sister visited me. (She) gave me several apples.

(2) 我久睡在一角的[席子]$_2$上, ϕ_2已经烤得那么热。
wo jiu shui zai yi jiao de xi zi shang, (ta) yi jing kao de na me re.
I was sleeping on a corner of the matting. (It) was heated up very hot.

(3) [这个问题]$_3$看上去很容易, 我们还没有解决ϕ_3。
zhe ge wen ti kan shang qu hen rong yi, wo men hai mei you jie jue (ta).
The problem seems to be very easy, but we have not resolved (it) yet.

(4) [他]$_4$起床ϕ_4穿好衣服。
(ta) qi chuang. (ta) chuang hao yi fu.
He got up and (he) dressed.

(5) ϕ_5墙上挂了一幅画。
(you ren) qiang shang gua le yi fu hua.
(Someone) put up a picture on the wall.

ZA resolution is important in many NLP tasks. One example is machine translation. For target languages such as English that cannot be adequately generated with omitted expressions, here the subject and the object, the antecedent of the ZA in the source language, must be identified and made explicit. Other examples include information-extraction and question-answering systems, where the relevant information about ZAs in text is difficult to find even when using large corpora. Recovering ZAs with their antecedents can improve the performance of such systems.

In this paper, we address ZA resolution in Chinese and argue that the proposed approach can be extended to other languages. In our work, we assume that the task of ZA detection has been performed by some other module, such as a shallow parser [3]. Our work focuses only on identifying the antecedent of ZA. We construct a maximum entropy (ME) classifier to check whether a candidate is the correct antecedent or not. In training and applying the classifier, we first employed a set of 13 regular features to capture the context information in discourses. From the original experi-

Manuscript received August 3, 2006.
Manuscript revised December 21, 2006.

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DOI: 10.1093/ietisy/e90–d.7.1092

*In the examples, we use ϕ_i to denote a ZA, where the subscript i is the index of its antecedent. The noun phrase attached with the same subscript is the antecedent of the ZA. Below the examples, we also give Chinese pinyin and English translation, in which the unexpressed elements are recovered and identified by ( ).

**In our work, independent clauses are viewed as sentences.

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mental results, we found that the classifier using the regular features mainly suffered from the problem of insufficient semantic knowledge. Therefore, we improved the classifier by developing two Web-based features to obtain additional semantic information from the Web. The values of the two features can be estimated by querying the Web. We combine the two features with the regular feature set and retrain the advanced ME-based classifier. In this way, we effectively incorporate the Web into our classifier and experimentally show that the Web, as a knowledge source, can improve the performance of our approach.

The rest of this paper is organized as follows: Section 2 gives an overview of previous research about ZA resolution. Section 3 describes the framework of the ME-based classifier, the data collection and data preparation. We present the 13 regular features and discuss their performance in Sect. 4. In Sect. 5 we extend the feature set with the Web-based features and evaluate the effectiveness of the advanced classifier. We compare our approach with related work in Sect. 6. Finally concluding remarks are given in Sect. 7.

2. Previous Research

To identify anaphoric relations, the existing research can be categorized into 3 classes: rule-based methods, statistical or machine learning models, and hybrid approaches.

In the first class, most approaches use heuristic rules to eliminate unacceptable candidates until the most likely antecedent is obtained. These hand-crafted rules were developed on the basis of real-world knowledge, pragmatics and a set of linguistic factors, which include gender, number agreement, semantic consistency, syntactic parallelism, salience, and proximity, for example [3]–[6]. However, writing a complete rule set is very labor-intensive and time-consuming work, and also requires linguistic expertise. Furthermore, if we want to take more factors and knowledge into account, it is difficult to satisfy the complicated heuristic rules.

In contrast to the labor-intensive rule-based methods, the second class employs statistical models [7], [8] and machine learning techniques [9], [10]. All the candidates of the antecedent are enumerated for an anaphora. In order to select the best candidate, each candidate is evaluated based on some features. For statistic models, the features can be the occurrence frequencies of a candidate in certain patterns, and for machine learning models the training features can include lexical, syntactic, semantic and positional information. The methods in this class rely heavily upon the availability of large corpora annotated with anaphoric relations. However, collecting such a training corpus also requires much work.

In order to alleviate the problem of much manual work being required, Isozaki and Hirao [11] proposed a hybrid approach, in which heuristic ranking rules and a machine learning system were combined, SVM. Their results showed that the combination made Japanese zero pronoun resolution more reliable. A similar hybrid architecture was employed by Hinrichs et al. [12], in which a rule-based prefiltering component was combined with a memory-based resolution module for anaphora resolution in German.

Although the approaches discussed above have made great contributions to anaphora resolution, most of them still have shortcomings. First, the test data in most experiments were selected from one single genre, such as newspaper articles or technical manuals. The problem of whether the proposed approaches are also effective for other genres was unexplored. The cataphora case is always ignored because it is assumed to occur infrequently. Finally, for Chinese text, the early work was carried out almost exclusively using rule-based approaches [3], [4], [6], and no attempt has been made to employ a machine learning framework. In our study, we attempt to improve the problems in the following ways. Instead of considering only one genre, we have used a diverse text corpus, consisting of newspaper articles, novels, and textbooks of different reading levels. Although cataphora occurs much less frequently than anaphora, we take it into account and develop some rules for it. In contrast to rule-based approaches, a classifier based on Maximum Entropy is employed in our resolution.

3. Our Approach for ZA Resolution

First, we deal with cataphora, in which the antecedents are found after ZAs. Cataphora occurs much less frequently than anaphora, thus, until recently, little attention had been given to it. In our study, we employ some simple heuristic rules for resolving cataphora because cataphora is very common in Chinese text. To extract the rules, we examined such cases in the corpus to identify the behaviour of cataphora in Chinese. The following steps are used to resolve cataphora examples with 100% accuracy in our corpus.

1. If a ZA appears in the first clause of a discourse, it is regarded as a cataphora.
2. Search the following sentences and stop at the first sentence that does not have a ZA.
3. If the subject of the sentence is not modified by a Named Entity (NE)† using the auxiliary word "de", the antecedent is the subject, otherwise the antecedent is the NE.
4. The antecedent(s) of the other ZA(s) between the cataphora and its antecedent is (are) the same as that of the cataphora.

The remainder of the paper focuses on the case where the antecedents come before ZAs.

3.1 ME-Based Classifier

Maximum Entropy (ME) is a general technique for estimating probability distributions from data. The general idea of ME is that the best probability model for the data is the one that maximizes entropy over the set of probability distribu-

†The module for NE recognition is described in Sect. 3.2.

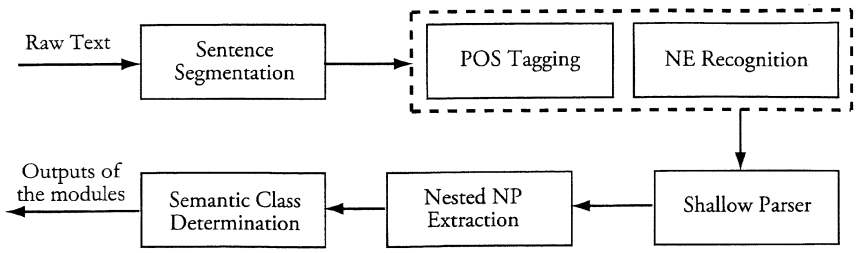


Fig. 1 System architecture of NLP modules.

tions that are consistent with events [13]. In the ME framework, events are represented as multiples of weighted features. In training, the weights of features are derived on the basis of the distribution of the features in the training data, using an iterative algorithm such as generalized iterative scaling (GIS) [14] or improved iterative scaling (IIS) [15]. Weights are chosen to maximize the entropy of the model while minimizing the divergence between the model and the training data. In employing the model, events of a given context are evaluated by summing the weights of their respective features and normalizing over the context to obtain a probability distribution, as in Eq. (1).

$$p(y|x) = \frac{1}{Z(x)} exp\left[\sum_i \lambda_i f_i(x,y)\right] \quad (1)$$

where $p(y|x)$ represents the probability of event y given context x, and λ_i represents the weight for feature f_i. $Z(x)$ is simply a normalizing factor to ensure a proper probability (Eq. (2)).

$$Z(x) = \sum_i exp\left[\sum_i \lambda_i f_i(x,y)\right] \quad (2)$$

Our use of the ME technique for antecedent identification is motivated by some important advantages of ME. First, it has been shown to perform well in combining heterogeneous forms of features and in handling interactions among features not suffering from any independence assumptions [16]. This allows us to take various features of antecedents and ZAs into account. ME can also work well when training material is limited [17] and this advantage allows us to reduce training expenses.

We develop a classifier based on ME to determine whether a candidate is the correct antecedent or not. Given a candidate $x \in$ {all possible candidates for a certain ZA}, we need to predict $y \in$ {yes, no}, which classifies x as the correct antecedent or not. The classification process starts from the noun phrase (NP) immediately preceding the ZA and proceeds backwards in the reverse order of the discourse until an antecedent is found. If the process exceeds the beginning of the discourse, we choose the NP in the previous sentence having the shortest distance from the ZA as its antecedent.

The primary work of modeling an ME-based classifier is the selection of a set of features (f_i) to be considered in the analysis of the training data and the estimation of the appropriate weight (λ_i) for each feature. In the latter work, we use IIS, a hill-climbing algorithm, to find the globally optimal weights given the training data. A complete description of IIS can be found in [15]. The features devised for ME modeling are described in Sects. 4 and 5.

3.2 Experimental Data

Instead of considering the ZAs of only one genre, that is, newspaper articles [3], [6], we have used a diverse text corpus, which is a collection of 85 text files from different genres, about 30 million words in all, consisting of newspaper articles (current affairs, politics, economy, sports and entertainment), Chinese fiction written by different authors and textbooks from elementary school to graduate school level. We randomly selected two subsets of the corpus as our training and test data. The data excludes three types of ZA: intrasentential, exophora and cataphora. The training data consists of 85 fragments, one fragment from each text file, with a total of 19,293 Chinese words. It includes 627 ZAs in all. There are 3,027 words and 121 ZAs in the test data. Raw text was input and automatically preprocessed by a pipeline of NLP modules, consisting of sentence segmentation, POS tagging, NE recognition, shallow parsing, nested NP extraction and semantic class determination. All possible annotations in a training text are determined by the outputs of these modules. The system architecture of the pipeline is shown in Fig. 1. We explain these modules as follows. In our work, the module of sentence segmentation segments an input text into sentences using three punctuation markers, which are "。" (period), "！" (exclamation mark) and "?" (question mark). These punctuation markers are sentential boundary marks in written Chinese. Our POS tagging and NE recognition were done on the basis of the Chinese lexical analysis system ICTCLAS[†], which was developed by the NLP group of Chinese Academy of Sciences. ICTCLAS is an approach based on multilayer HMMs and it includes word segmentation, POS tagging and NE recognition. The segmentation precision of ICTCLAS is 97.58% and the recall rates of POS tagging and NE recognition are higher than 90% (the results from recent official evaluation in the national 937 project[††]).

[†]http://sewm.pku.edu.cn/QA/reference/ICTCLAS/FreeICTCLAS.

[††]A project supported by the National Natural Science Foundation of China and the China National Key Programme for Development of Basic Sciences.

Table 1 Summary of regular features (RFs).

Type	No.	Feature	Description	Values
Grammatical	1	PREDI-ZA	Surface form of the head of the predicate in ZA-S.	VP, AP, PP, MP, NP, unknown
	2	PREDI-A	Surface form of the head of the predicate in A-S.	VP, AP, PP, MP, NP,unknown
	3	NESTED-A	The candidate is a nested NP.	yes, no
	4	GRAM-FUNC	The grammatical role of the candidate.	subject, object, embedded subject, embedded object, attributive, adverbial modifier, others
Lexical	5	P-P	Lexical relation of the heads of the predicates in ZS-S and A-S.	same, synonym, antonym, others
	6	DEMON-A	The candidate starts with a demonstrative pronoun.	yes, no
	7	PRON-A	The candidate is a pronoun.	yes, no
	8	NE-A	The candidate is a NE.	yes, no
	9	CONJ	Type of conjunction between the candidate and the ZA.	coordinate, continued, causality, transitional, others
Positional	10	DIST	Distance between the candidate and the ZA in simple sentences.	1, 2, 3, 4
Semantic	11	SEM-A	Semantic class of the candidate person.	animal, plant, object, artifact, space, time, data, process ... unknown
	12	SEM-P	Semantic class of the head of the predicate in ZA-S.	state, emotion, change, weather, perception, motion, communication ... unknown
	13	SEM-P-A	Semantic class of the head of the predicate in A-S.	state, emotion, change, weather, perception, motion, communication ... unknown

ZA-S: the sentence containing the ZA; A-S: the sentence containing the antecedent candidate
VP: verb phrase; AP: adjective phrase; PP: prepositional phrase; MP: numeral-classifier phrase

Our shallow parser† uses the approach proposed in [3], which parses a sequence of words into smaller constituents such as NP and verb phrases (VP) with phrase-level parsing. The module of nested NP extraction divides nested NPs into two groups.

1. Nested NPs that are modifier nouns. For example, the nested NP for "shi yan jie guo (experimental result)" is "shi yan (experiment)".
2. Nested NPs that modify the heads using the auxiliary word "de (of)". For example, for "er tong de jiao yu (children's education)", the nested NP is "er tong (children)".

Semantic classes of phrases are determined based on the Semantic Knowledge-base of Contemporary Chinese (SKCC) [18]. We defined 9 semantic classes for NPs and 7 classes for VPs. We assumed that the semantic class for each phrase is the most frequent sense of its head.

Each ZA was extracted and considered in a 4-sentence context. The correct antecedents were manually annotated. In the training data, each pair of a ZA and its antecedent is considered to be a positive training instance for our classifier, while all the other NPs occurring in the scope between the ZA and it antecedent form negative training instances with the ZA. These NPs consist of general NPs, nested NPs and NEs. This procedure produced a set of 3,261 training instances, in which 627 (19.2%) were positive and 2,634 (80.8%) were negative. In the test data, all the NPs extracted in the context for a certain ZA are regarded as possible antecedent candidates and are used in the later classification experiments.

4. The Classifier with Regular Features

In this section, we describe our ME-based classifier using a set of regular features. In designing the feature set, special attention was paid to features that were (1) relevant according to previous research [9]–[11], (2) cheap and easy to annotate automatically, and (3) robust and domain-independent. This resulted in the 13 features shown in Table 1. Although these features might not always work best for a specific genre compared with state-of-the-art systems [3], [4], they are considered to be robust and applicable to diverse texts. The values of the features can be obtained directly from the corpus, from the pipeline of NLP modules, e.g., our POS tagger and shallow parser, and using semantic dictionaries with the help of heuristic rules.

4.1 Regular Features

The set of 13 regular features (RFs) includes 4 grammatical, 5 lexical, 1 positional and 3 semantic features.

Grammatical features. PREDI-ZA and PREDI-A are the surface forms of the heads of the predicates in the sentences containing the ZA and the antecedent candidate. The values come from the POS tagger and shallow parser. The fea-

†There are several reasons why we employ a shallow parser rather than a traditional full parser. Full parsers aim at complete, exact parses. Thus they do not do well in identifying good phrases in noisy surroundings. However, diverse texts and the occurrence of ZAs unavoidably introduce noise to full parsing. On the other hand, shallow parsers aim at analyzing syntactic information efficiently and reliably for unrestricted text with ellipsis of grammar roles. Thus they are useful for our task.

ture NESTED-A checks whether the antecedent candidate is a nested NP or not. The feature GRAM-FUNC shows the grammatical role of an antecedent candidate and the values are obtained using our shallow parser with some heuristics.

Lexical features. The feature P-P describes the lexical relation between the heads of the predicates in the sentences containing the ZA and the candidate. The four possible values can be obtained based on a synonym/antonym dictionary. The feature DEMON-A checks whether the candidate starts with a demonstrative pronoun or not. PRON-A and NE-A judge whether the candidate is a pronoun or a NE. The feature CONJ evaluates the type of conjunction between the ZA and the candidate. The values can be determined directly by the conjunctions occurring in the sentences. For example when a conjunction, such as "bing qie (and)", "bing (and)" and "tong shi (at the same time)", occurs in the sentence, the value is "coordinate". When the conjunction is "ying wei (because)" or "suo yi (so)" or "zhi suo yi (the reason why ...)", the value is set to be "causality".

Positional feature. The feature DIST measures the distance between the antecedent candidate and the ZA in numbers of simple sentences.

Semantic features. The features SEM-A, SEM-P and SEM-P-A describe the semantic classes of the antecedent candidate and the heads of the predicates in the sentences containing the ZA and the candidate, respectively. All the values of these three features can be determined using a Chinese semantic dictionary.

The values of the features cannot always been obtained from the pipeline of NLP modules directly. Four features, GRAM-FUNC, DEMON-A, CONJ and DIST, require the further help of heuristics. In our work, we developed 40 rules based on the outputs of the NLP modules to determine the values of these features. For example, in preprocessing the raw text, sentences were segmented using the three punctuation marks. In determining the value of DIST, the compound sentences between ZA and the antecedent candidate must be further divided into simple sentences. One of the simple sentence rules is "If a dependent clause† starts with a conjunction word, such as 'dan shi (but)', 'bu guo (but)', 'bing qie (and)' and 'ran hou (then)', the clause is viewed as a simple sentence."

4.2 Experimental Results and Error Analysis

In order to show the improvement of the performance of our approach, we developed the following simple rule-based baseline algorithm.

Baseline rule:

For each ZA in a sentence, choose the NP in the previous sentence having the shortest distance from the ZA as its antecedent.

Table 2 shows the results for the ME-based classifier using RF in comparison with the baseline on the test data (121 ZAs). The performance is evaluated as to *accuracy*, that is, the number of correctly resolved ZAs divided by the total number of ZAs.

Table 2 Results with RFs.

Feature	Baseline	RF
Correct number	56	87
Accuracy	46.3%	71.9%

The baseline successfully resolved 56 ZAs in our test data. Unlike the ME-based classifier with RF, the baseline is not limited by the performance of the preprocessing NLP modules or the quality of the semantic resources, and therefore, it resolved some of the samples (4 in total) that the classifier did not resolve, e.g., example (6) below. The antecedent for ϕ_6 determined by the baseline is "si xiang dou zheng (the struggle)" (actually it is) while the antecedent determined by our approach is "xi (play)".

(6) 这场戏向观众揭示出了人们内心深处的[思想斗争]$_6$, ϕ_6既想履行责任, 但又顾虑重重。
zhe chang xi xiang guang zhong jie shi chu le ren men nei xin shen chu de si xiang dou zheng, (si xiang dou zheng) ji xiang lv xing ze ren, dan you gu lv chong chong.
The play represents the inner struggle in a man's mind to the audience. (the struggle) is always between duty and doubt.

On the other hand, there are many examples where our approach was correct and the baseline was incorrect, e.g., examples (7), (8) and (9) below. The main reason for failure was that the baseline ignored many factors, such as grammatical information and semantic knowledge, which play roles in ZA resolution.

(7) [北京鸭]$_7$是一种中国种的大白鸭。ϕ_7在美国广泛培育用于食用。
Beijing ya shi yi zhong zhong guo zhong de da bai ya. (ta) zai mei guo guang fan pei yu yong yu shi yong.
Beijing duck is a large white duck of a Chinese breed. (It) is widely raised in the United States for food.

(8) [她]$_8$感到心绪烦乱, 事情发生得如此之快, ϕ_8不知下一步如何是好。
ta gan dao xin xu fan luan, shi qing fa sheng de ru chi zhi kuai, (ta) bu zhi xia yi bu ru he shi hao.
She is feeling very worried. Things are happening so fast. (She) does not know what to do next.

(9) 为了让自己有时间思考, 我把[烟丝]$_9$轻按几下塞紧在烟斗里, 然后我慢慢地点着ϕ_9。
wei le rang zi ji you shi jian si kao, wo ba yan si qing an ji xia sa jin zai yan dou li, ran hou wo man man de dian zhao (ta).
To give myself time to think, I tamped down the tobacco in my pipe and then lit (it) slowly.

†Dependent clauses are signaled by having a conjunction, such as "although" and "because", or a relative pronoun, relative adjective, or relative adverb, such as "that", "which", and "where".

Table 3 Classes and numbers of errors.

Type	Number	%
Errors in POS tagger	3	8.8%
Errors in NE recognition	1	2.9%
Errors in shallow parser	4	11.8%
Word sense ambiguity	3	8.8%
Errors of heuristic rules	7	20.6%
Lack of semantic knowledge	16	47.1%

Our approach performs much better than the baseline. While these results are encouraging, there were several classification errors. The six major types of errors and their frequencies are summarized in Table 3. The first three types of errors are caused by mistakes made by the POS tagger, the NE recognition and the shallow parser. These modules may output incorrect values for the regular features, causing classification errors (23.5% of all errors). The problem of how to improve the performance of these modules is not considered in our work and is beyond the scope of the paper. Word sense ambiguity is another source of misclassification. We use SEM-A, SEM-P and SEM-P-A to recognize the semantic classes of the candidate and the heads of the predicates with the help of a Chinese semantic dictionary. The semantic class determination module assumes that the semantic class for every phrase is the first sense of the head noun of the phrase, that is, it always chooses the most frequent sense as the semantic class. However, one word may have several word senses and its sense class in a certain sentence should be determined by the context. Word sense ambiguity makes up 8.8% of all errors.

The heuristic rules used for determining the values of regular features cause 20.6% of all errors. This is due to cases that are not covered by the rules. For example, in determining the distance between the current candidate and the ZA, the phrase that starts with a conjunction, e.g., "bu guo 10 mi (just 10 meters)", was mistakenly viewed as a simple sentence.

The largest proportion of errors is errors due to the lack of semantic knowledge (47.1%). The examples of this type of errors can roughly be classified into two groups.

1. Examples having word senses that are missing from the semantic dictionaries, e.g., "second quarter" should be assigned the semantic class "data" instead of "unknown" in example (10) below.

2. Examples that require general world knowledge beyond simple word meanings, e.g., examples (11) and (12) below. In (11), ϕ_{11} and ϕ_{12} can be interpreted correctly using the knowledge that bananas, being a fruit, are likely to be unripe and the monkeys, being an animal, are likely to be hungry. In (12), ϕ_{13} refers to "Li Si" because "Zhang san" no longer has the apple. The general world knowledge on "give" is that the actor no longer has the object after giving it to others.

(10) 每年的[第二季度]$_{10}$收入都会减少, ϕ_{10}一直是PC界最差的销售季度。
mei nian de di er ji du shou ru dou hui jian shou, (di er ji du) yi zhi shi PC jie zui cha de xiao shou ji du.
Revenue decreases in the second quarter of each year. (The second quarter) is always the slowest quarter of the year for the PC industry.

(11) 他掏出准备好的[香蕉]$_{11}$, 虽然ϕ_{11}还没熟, 他把香蕉给[猴子]$_{12}$, 因为ϕ_{12}实在是饿了。
ta tao chu zhun bei hao de xiang jiao, sui ran (xiang jiao) hai mei shu, ta ba xiang jiao gei hou zi, yin wei (hou zi) shi zai shi tai e le.
He took out the prepared bananas. Although (they) were not ripe, he gave them to monkeys. Because (the monkeys) were very hungry, they hurried up and scrambled for them immediately.

(12) 张三给[李四]$_{13}$一个苹果, ϕ_{13}很快就吃了。
Zhang san gei Li si yi ge ping guo, (Li si) hen kuai jiu chi le.
Zhang San gave Li Si an apple. (Li Si) ate the apple.

From the error analysis above, it is obvious that our approach mainly suffers from semantic problems, such as semantic ambiguity and the lack of semantic knowledge. To our knowledge, these problems cannot be resolved using any current semantic dictionary.

5. The Advanced Classifier with Web-Based Features

Corpus-based approaches usually suffer from a data sparseness problem even in a very large corpus. The Web has a vast amount of data and the volume is much larger than that of any normal corpus. Therefore, the Web is increasingly being used as an information resource in a wide range of NLP tasks, such as machine translation [19], question answering [20] and language modeling [21]. Lapata and Keller [22] describe how to use Web-based models for resolving eight NLP tasks and show the promising performance of those unsupervised systems. There have also been efforts to employ the Web for anaphora resolution. Markert et al. [23] proposed an unsupervised method to deal with non-pronominal anaphora resolution using simple Web queries with shallow linguistic patterns. Two kinds of anaphora, other-anaphora and bridging in English text are addressed in their work. Motivated by this background, in our work, we attempt to use the Web as an additional semantic knowledge corpus for ZA resolution by combining the Web and our ME-based classifier.

5.1 Using the Web as a Knowledge Corpus

We employ the Web as a semantic knowledge resource on the basis of two facts discovered by careful examination of the experiment examples. One fact in ZA resolution is that the semantic constraint that is satisfied by a ZA, must be satisfied by its antecedent. That means that for a certain ZA, its antecedent and predicate must have semantic consistency. For example, considering ϕ_{11} and ϕ_{12} in (11), the antecedent of ϕ_{11} should be "bananas" because the candidate "bananas"

has higher semantic consistence than "monkeys"[†]. For the same reason, the antecedent of ϕ_{12} should be "monkeys" not "bananas" because the former has higher semantic consistence. The other fact is that although the relation between a ZA and its antecedent is implicitly expressed, they are linked by a strong semantic relation between the heads of the predicates of the sentences containing the ZA and its antecedent, denoted as V_{antec} and V_{ZA}. For example, considering ϕ_{13} occurring in the subject position in (12), the head pair [give :: eat] has the *transfer* relation, which indicates two actions are conducted by different subjects, that is, the subject is supposed to change from one sentence to the next. Therefore the antecedent of ϕ_{13} is not the subject of the previous sentence, "Zhang san", but "Li Si". Examples of head pairs of the *transfer* relation include [show :: see], [fell :: fall], and [give :: have]. Besides the *transfer* relation, another relation, the *happens-before* relation defined between V_{antec} and V_{ZA} is considered in our work. This type of relation indicates that the two disjointed actions are conducted by the same subject. Examples of this type of relation include [marry :: divorce], [enroll :: graduate], and [get up :: dress].

From the points discussed above, it is obvious that the semantic consistency and the semantic relations between predicates could benefit ZA resolution. We compute the semantic consistency and discover the semantic relations by querying the Web with certain patterns indicative of each relation. Although querying the Web adds noise to the data and we certainly lose some precision compared with supervised statistical models, we assume that the vast size of the Web will compensate for the rough queries. We extract the Web knowledge by the following procedure.

1. **Construct lexico-syntactic patterns**. To compute the semantic consistency, two kinds of predicate-argument relations, subject-predicate and predicate-object, are considered in our work. The patterns are constructed in the form of "NP VP" (for subject-predicate) and "VP NP" (for predicate-object). The patterns for discovering the semantic relations between V_{candi} and V_{ZA} are manually designed and refined by examining examples in a volume of known semantic relations. For example, to detect the *happens-before* relation, the patterns used could be "xian V_{candi} zai V_{ZA} (V_{candi} and then V_{ZA})" and "V_{candi} jie zhe V_{ZA} (V_{candi} and later V_{ZA})". To detect the *transfer* relation, the patterns could be "bei V_{candi} ran hou V_{ZA} (be V_{candi} ed and then V_{ZA})" and "bei V_{candi} jie zhe V_{ZA} (be V_{candi} ed and subsequently V_{ZA})". In our work, we use a total of 17 patterns for extracting semantic knowledge from the Web.
2. **Instantiate patterns**. We instantiate the patterns for all antecedent candidates. For example, for ϕ_{11} in (11), the pattern "NP VP" is instantiated with "the bananas are ripe" and "the monkeys are ripe". If a candidate is a pronoun, it would be replaced by its closest nominal antecedent before instantiation. These instantiated patterns are submitted to a Web search engine (Google[††] in our work). All queries are performed as exact matching. Sometimes even the Web is not large enough to contain the instantiated pattern and the data sparseness problem still exists. Therefore, some processing is necessary to reduce data sparseness before instantiated patterns are submitted to Google. First, all modifications in sentences are deleted and only the heads are retained in the patterns. For instantiated patterns containing a NE, the NE should be replaced by the common noun that corresponds to the semantic class of the NE, e.g., "China" is replaced by "country".
3. **Acquire semantic knowledge from the Web**. To compute the semantic consistency from the Web, our rationale is that if the consistency between the ZA and a candidate is high, it is likely that the pattern instantiated by the candidate occurs frequently on the Web. We use *ConsiScore* to estimate semantic consistence using Eq. (3), where *count(P)* is the Web count of the instantiated pattern returned by Google, while *count(candi)* is the Web count of the query formed with only the head of the current candidate. N is the number of Google pages ($N \approx 10.1 \times 10^9$). To determine the semantic relations, we adopt a method inspired by mutual information (MI) to measure the strength of association between V_{candi} and V_{ZA}, denoted $R_P(V_{candi}, V_{ZA})$, which is estimated using Eq. (4), where $count(V_{candi})$ and $count(V_{ZA})$ are the Web counts of the queries formed by V_{candi} and V_{ZA}. We determine that the semantic relation R_P indicated by pattern P is present between V_{candi} and V_{ZA} if $R_P(V_{candi}, V_{ZA})$, exceed a certain threshold θ[†††].

$$ConsiScore(candi, ZA) = \frac{\frac{count(P)}{N}}{\frac{count(candi)}{N}} \quad (3)$$

$$Rp(V_{candi}, V_{ZA}) = \frac{\frac{count(P)}{N}}{\frac{count(V_{candi})}{N} \times \frac{count(V_{ZA})}{N}} \quad (4)$$

5.2 Combing the Web and the ME-Based Classifier

Unlike research in which web counts are used directly [22], our method of employing the semantic knowledge from the Web is to combine the Web knowledge as features together with the 13 RFs described in Sect. 4.1. The two Web-based features with their possible values are summarized in Table 4. The feature SEM-CONSI obtains the semantic consistency computed from the Web. The values represent the rank of the consistency. For example, SEM-CONSI = 1 means the current candidate has the highest semantic consistency among all the antecedent candidates. The feature

[†]We can say "the bananas are not ripe" but never say "the monkeys are not ripe".

[††]http://www.google.com

[†††]$\theta = 0.3$ in this paper. We adjusted the value of θ by performing exhaustive search on a small annotated corpus, that is, all possible values of θ ($0 \le \theta \le 1$) were tried with a step size of 0.1.

Table 4 Two Web-based features.

Type	No.	Feature	Describe	Values
Web-based	14	SEM-CONSI	The rank of the semantic consistency of antecedent candidates.	1, 2, 3, 4 ...
	15	SEM-RELA	The semantic relations between the heads of the predicates in sentences containing ZA and an antecedent candidate.	transfer, happens-before, unknown

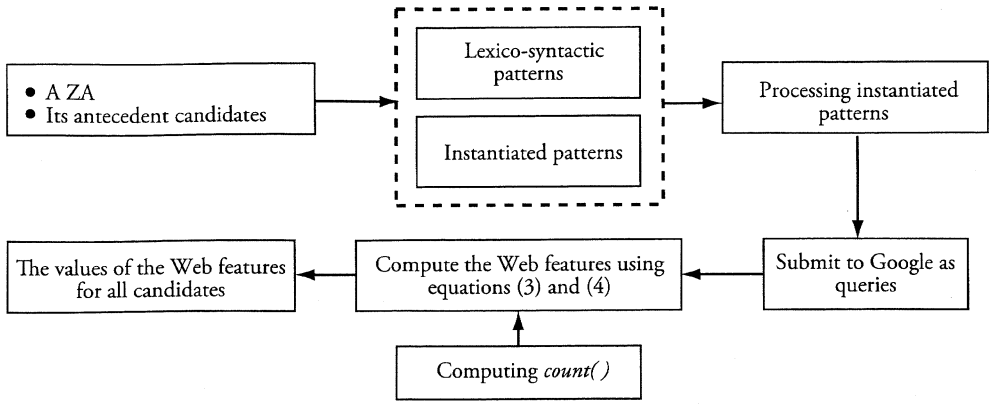


Fig. 2 Process flow of computing the values of Web-based features.

SEM-RELA represents the semantic relations between the heads of the predicates of ZA and a candidate, as described in Sect. 5.1. In our current work, three values are possible: "transfer", "happens-before" and "unknown".

Figure 2 shows the process flow for computing the values of the Web-based features of all candidates for a ZA. For a certain ZA, suppose we compute the values of the two Web-based features for all candidates. Following the procedure described in Sect. 5.1, we first instantiate the developed lexico-syntactic patterns for each candidate. We then obtain instantiated patterns. In order to mitigate data sparseness, the patterns are processed again before they are submitted to Google as queries. Finally, we determine the values of the Web-based features using Eqs. (3) and (4). In computing the Web counts of each query (*count( )*), we did not use the Google Web API[†] in our program implementation because although Google provides users various applications, it limits the amount of automated queries to 1,000 per day. We developed our extraction tool in C++. The tool merely extracts Web count numbers from the Web pages directly and thus has no restrictions on the number of search queries.

5.3 Experimental Results

We retrained and tested the advanced classifier with the RF set, and the two additional Web-based features. The results are summarized in Table 5. The right-most column shows the performance of the advanced classifier. We obtain 35.5 and 9.9 point improvements in accuracy in comparison with the baseline and the original classifier trained on the RF set. In particular, all the examples given in this paper can be resolved by the advanced classifier.

RF+Web found more correct antecedents than RF, achieving an 81.8% accuracy vs 71.9%. In particular, RF+Web was more successful than RF in checking the semantic consistency that is always inadequate in normal semantic dictionaries. For example, it correctly resolved example (11), for which RF had determined "ta (he)" as the correct antecedent for both of the two ZAs mistakenly. It can also correctly resolve the following example (13) for which RF had determined "mu qin (mum)" mistakenly.

Table 5 Results with RF and RF+Web-based features.

Feature	Baseline	RF	RF+Web
Correct number	56	87	99
Accuracy	46.3%	71.9%	81.8%

(13) 母亲高兴地答应了, 拿了篮子, 把木槿[花]$_{14}$全都摘下来了。廖大夫拿秤一称ϕ_{14}, ϕ_{14}竟有一斤。
mu qin gao xing de da ying le, na le lan zi, ba mu jin hua quan dou zhai le xia lai. Liao dai fu na chen yi cheng (hua), (hua) jing you yi jin. Mum agreed with great pleasure.
She took out the basket and picked off all the flowers of hibiscus. Dr. Liao weighed (the flowers), which was actually about 500 grams.

RF+Web had a higher success rate with examples that required implicit semantic relations between ZA and its antecedent to be identified. For example, it correctly resolved example (12), for which RF had mistakenly determined "Zhang san" as the correct antecedent. It can also correctly resolve the following example (14) for which RF had mistakenly suggested "di di (younger brother)".

(14) [我]$_{15}$走下楼, ϕ_{15}吃过早餐, 尽管弟弟昨天给我开了个不大不小得玩笑, ϕ_{15}还是给这家伙打了个招

[†]Google offers open Web Services with 1,000 searches per day. Using Google Web API, users can have access to real-time data (such as Google News, Google Images, and Google Book Serach) through a simple, fast, and easy-to-use search interface.

Table 6 Performance of leave-one-out classifiers and leave-group-out classifiers.

No.	Feature	Correct number	Accuracy (%)	Group	Correct number	Accuracy (%)
1	PREDI-ZA	99	81.8			
2	PREDI-A	99	81.8	Grammatical	89	73.6
3	NESTED-A	99	81.8			
4	GRAM-FUNC	91	75.2			
5	P-P	99	81.8			
6	DEMON-A	99	81.8			
7	PRON-A	98	81.0	Lexical	93	76.9
8	NE-A	98	81.0			
9	CONJ	97	80.2			
10	DIST	100	82.6	Positional	100	82.6
11	SEM-A	99	81.8			
12	SEM-P	99	81.8			
13	SEM-P-A	99	81.8	Semantic	85	70.2
14	SEM-CONSI	92	76.0			
15	SEM-RELA	97	80.2			
	Baseline	56	46.3	Baseline	56	46.3
	All features	99	81.8	All features	99	81.8

呼, ϕ_{15}心情愉快地出门了。
wo zou xia lou, (wo) chi guo zao can, jing guan di di zuo tian gei wo kai le ge bu da bu xiao de wan xiao, (wo) hai shi gei zhe jia huo da le ge zhao hu, (wo) xin qing yu kuai de chu men le.
I went down the stairs and had my breakfast. Although my younger brother played a joke on me, I said hello to him. And then (I) went out.

Although our approach has been improved greatly, some test examples are still misclassified. There are several reasons for this. First, the mistakes in pre-processing modules (POS tagger, NE recognizer and shallow parser) inevitably affect the performance of the classifier. Another problem is the limitations of Web counts, which lie in the introduction of noisy data from the Web and the instability of Web search engines.

5.4 The Contribution of the Features

It is interesting to find out how useful each of our 15 features is in the ZA resolution. Our way to do this is to train and test a series of leave-one-out classifiers where we disabled one feature at a time (Nos. 1-15). The contributions of the features are measured in terms of the changes of ZA resolution accuracy. We assume that when a certain feature is disabled, if the accuracy decreases significantly, the feature makes a significant contribution. If the accuracy does not change, the feature is not important. If the accuracy increases, the feature has a confusing effect and it should be removed. The performance of these leave-one-out classifiers is shown on the left side of Table 6. The features that cause a substantial reduction of accuracy when disabled are GRAM-FUNC and SEM-CONSI. The remaining features only have a small impact on the overall accuracy. The classifier with all the features performs the best, that is, when no feature is disabled, the accuracy is the highest.

To evaluate the joint contribution of similar knowledge sources, e.g., grammatical and semantic features, we ran four additional leave-group-out classifiers where we disabled the features of one group at a time[†]. The performance of these classifiers is shown on the right side of Table 6. The semantic features play the most important role in ZA resolution and the group of grammatical features is the second most important.

6. Comparison with Related Work

Although there have been many efforts at ZA resolution in Chinese, most of them have not been subjected to empirical evaluation until recently. In this section, we compare our work with two approaches proposed by Yeh and Chen [3] and Zhang and Zhou [4]. We will consider two aspects, the principles of these approaches and the performance with the same test data.

Zhang and Zhou [4] proposed a rule-based approach for ZA resolution by investigating volumes of authentic examples from corpora. The handcrafted rules are used as filters to discount unacceptable candidates rather than to propose antecedents. However, when applying these rules, we find that some of them are not computable. For example, to our knowledge, no algorithm so far can distinguish between the topic and subject in Chinese discourse. The two grammatical roles are very commonly seen in the rules for ZA resolution. In our study, we employed a machine learning approach to the task. Our idea was to cast ZA resolution as a classification problem for which a classifier can be automatically trained using a certain learning method on the annotated corpus. We evaluated the performance of the rules proposed in [4] for the same test data as used in our work (Sects. 4.2 and 5.3). The accuracy was 76.0% (92 of 121 ZAs), while that of our approach was 81.8% (99 of 121 ZAs).

Yeh and Chen [3] adopted Centering Theory (CT) [5] as the framework for ZA resolution. Although the syntactic and semantic criteria based on CT for the selection of an antecedent are very strong, they are not always suf-

[†]The two Web-based features are grouped into the semantic group.

ficient to distinguish between possible candidates, and the features employed in their approach have proved to be insufficient to achieve a satisfactory performance. Only the positions and the grammatical roles of antecedent candidates are considered. Therefore, these simple heuristics are not able to perform as intelligently in other genres (e.g., fiction) as in newspaper articles. In our study, we considered those domain-independent features that could be obtained at low cost but with high reliability. The features used were of two groups, regular features and Web-based features. The selection of these regular features had theoretical backgrounds. For example, our feature set included the grammatical function and position-related features derived from research on CT. In order to improve the performance of our approach further, we also employed two Web-based features to extract additional semantic knowledge from the Web. We evaluated the performance of the approach in [3] for the same test data as used in our work. The accuracy was 62.8% (76 of 121 ZAs), while that of our approach was 81.8%.

In all of the performance comparisons described above, our approach was shown to perform best. The contribution of our work lies in showing that a machine learning approach is able to achieve accuracy competitive with that of state-of-the-art systems using rule-based approaches. To solve the problem of insufficient semantic knowledge that is not encoded in any current semantic dictionary, we utilized information from the Web and combined it with other features in our machine learning framework. Our approach was shown to be simple and effective. Finally, our work was also the first in which various types of text were used as test data. Besides newspaper articles, which are used in most of the related work, our test data also included fiction and textbooks of different reading levels.

7. Conclusion

In this paper, we presented a machine learning approach based on ME for ZA resolution in Chinese. Besides the commonly used features, we developed two innovative Web-based features to obtain additional semantic knowledge from the Web. The values of the two features can be estimated by searching the Web using some simple patterns. Our study shows that the semantic knowledge obtained from the Web can be effectively introduced into our machine learning framework and can significantly improve the performance of ZA resolution.

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